

Adham Amgad

Industrial Automation / Control Systems Engineer

adham.amgad0710@gmail.com | 01210104535 / 01553339144 | Gesr Elsuez, Cairo, Egypt

LinkedIn: www.linkedin.com/in/adham-amgad1

PROFILE

A motivated and detail-oriented Electrical Power and Machines Engineer with ~2 years of experience in Industrial Automation, PLC programming, and DCS systems, worked on cement and food&beverage process. Skilled in P&ID interpretation, PID loop tuning, and field instrumentation including sensors and transmitters. Experienced in Siemens PCS7 (DCS), SCADA systems (Ignition V8.1, WinCC), and industrial communication protocols such as ProfiNet, OPC UA, Profibus, and Modbus RTU. Passionate about Industry 4.0, SAS, and driving digital transformation in industrial environments.

PROFESSIONAL EXPERIENCE

Controls Systems Engineer | Controware – Zahraa Elmaadi 06/2026– Present

- Process P&ID reading and interpretation for irrigation project in Saudi Arabia
- RTUs' I/O List and SCADA mapping list Verification for irrigation project in Saudi Arabia

Controls /Field service Engineer |JBT Marel (Part-time) – Relocatable 05/2026– 06/2026

- Working on electrical panel wiring revision and checking for food&beverage process.
- Working on site commissioning for PLC Program TIA Portal &Rockwell Rs-Logix 5000.
- Working on HMI edits based on client request and needs using FactoryTalk View SE.
- Working on Configuration of routers including “Ewon Flexy 205” for IIOT integration.

Industrial Automation Engineer | IMHOSYS Company – Maadi, Egypt 09/2024 – 05/2026

- Upgrade of Sinai Cement Grey Co. from Allen Bradley to PCS7 V9.1 based on CEMAT Minerals Library.
- WinCC SCADA Design, Logic Design with FBD using CFC tool, and Bulk Engineering Creation using IEA.
- Configured various Siemens PLCs and processors including S7-400, ET200SP, ET200SP HA, CPUs 1500
- Performed Site Survey, FAT/SAT Testing, I/O testing, calibration verification, and Site Commissioning.
- Understood and implemented Network Architecture linking servers, clients, Level 1 stations, and Engineering stations.
- Worked on Automation Material Selection and network topologies.

Maintenance Engineer | ALASHRAM HI ONE Textile Group – 10th of Ramadan07/2024 – 08/2024

- Control panel wiring, electrical installation of a new RAM Textile production line.
- Troubleshooting electrical faults and supporting MCC inspection during production line setup.

PROJECTS

- **Green Riyadh Project – KSA(2026)** :Working on P&ID Interpretation and IO list and Mapping List verification for RTU and SCADA.
- **Arabia For Cement – Ain Sokhna (2026)**: Extension of packing area including drawing mimics in ECS SCADA and logic modification by Step7.
- **Pyrometals Industries – KSA (2026)**:Metal powder (slag grinding) process: PCS7 V10 DCS project creation, logic design, P&ID/flowsheet reading, equipment & instrument list review, and system architecture and topology design.
- **Lemon and orange Squeezing machines–Egypt (2026)**:ProfiNet troubleshooting,electrical panel wiring revision,PLC logic test,Sequence troubleshooting and Allen-Bradely HMI Modification.
- **Sinai Cement L1 Upgrade (2024-2025)**: Working on 3 phases of a Mega Project upgrade consisting of 9 PLCs, I worked on 6 of them from Legacy allen Bradely to PCS7 V9.1 DCS:project creation, PLC programming, SCADA design, commissioning of 3 PLCs S7-400, FAT/SAT, I/O testing, P&ID reading, ProfiNet/Profibus/Modbus RTU interfaces.Delivered technical training sessions to Maintenance Engineers on troubleshooting.

TECHNICAL COURSES

- Substation Automation (SAS): IEC 61850 & IEC 104 - From Zero to Real Projects 06/2026
- SAS | Deep Dive and Roadmap – E/Hossam Embaby (Udemy) 05/2026
- Ignition 8.1 and 8.3 SCADA Credential Course – Inductive University 05/2025 – 04/2026
- ISA-88 PLC Programming Standard – E/Ahmed Saleh (Udemy) 08/2025
- HA Consulting Group:
 - ▶ Advanced Industrial Automation Diploma: SCADA Design, Advanced PLC, Communication Protocols (ProfiNet, Modbus RTU, Profibus) | 09/2023 – 11/2023
 - ▶ Basic Industrial Automation Diploma: Basic PLC, Classic Control, and Motor Drives | 08/2022 – 09/2022

EDUCATION

Bachelor of Engineering – Electrical Power and Machines | Zagazig University 09/2019 – 07/2024

- Accumulative Grade: Very Good (77.54%)
- Graduation Project: Smart Operation of Induction Motor Drives fed from PV Source Using PLC and IIoT – Grade: Excellent (200/200)

TRAININGS & INTERNSHIPS

- IMHOSYS Company – Process Automation Internship 07/2024 – 09/2024
- ArabTech for Electrical Industries 08/2023 – 09/2023
- ABB – MyLearning-MyFuture 2023 Internship 07/2023 – 08/2023
- ELSEWEDY ELECTRIC – The United Wires Branch 08/2022 – 09/2022

PROGRAMMING & TECHNICAL SKILLS

SCADA: Ignition V8.1 & 8.3, WinCC Explorer, ECS SCADA, Wonderware InduSoft v8.0, WinCC Unified

HMI: WinCC Comfort TP900(Siemens), Vejo Designer(Schneider), DOPsoft (Delta), FactoryTalk View SE(Rockwell)

Database: Microsoft Office; Beginner in MS SQL Server Studio & MySQL

IoT: Configure Ewon Router, Beginner in NodeRed – OPC UA & MQTT integration with PLCs

PLC: Siemens TIA Portal, PCS7 DCS (Ladder, SCL, STL, FBD, CFC); Allen Bradley RS-Logix 5/500/5000.

RTUs and IEDs(SAS) :Easergy Builder,eSetup Easergy Pro, CET850,OMICRON IEDScout

Motor Control: VFD & power monitor parameter setting;Modbus RTU (RS-485) communication

Communication: ProfiNet configuration (SINEC PNI); Profibus, OPC UA, Modbus RTU, IEC61850 and IEC104.

Network: SINEC PNI, Wireshark ,Ecatcher ,Ebuddy

Design: Classic control circuit design; AutoCAD Electrical & EPLAN fundamentals

High-Level Languages: C (Good – Academic), Java (Good – Academic), Python (Beginner – Self-study)

NON-TECHNICAL COURSES

- ALX Africa – AICE Program 05/2024 – 07/2024
- PMI – Kickoff Program (Fundamentals of Predictive Project Management) 06/2024
- McKinsey Academy – Step Forward Program 10/2023 – 04/2024
- Aspire Training Solutions – Employability Skills 07/2022 – 08/2022

EXTRA-CURRICULAR ACTIVITIES

- Basic Industrial Automation Track Lead – IEEE Zagazig Student Branch (PES) | 10/2023 – 04/2024
- Project Management Trainee – VEP (Volunteering Empowerment Program) | 07/2023 – 08/2023
- Technical Support Member – IEEE ZSB (PES) | 02/2023 – 10/2023
- Industrial Automation Member – IEEE ZSB (PES) | 02/2022 – 02/2023

LANGUAGES & SOFT SKILLS

Languages: Arabic (Native) | English – Upper Intermediate (B2) | German – Beginner (A1)

Soft Skills: Adaptability • Teamwork • Attention to Detail • Time Management • Problem Solving • Communication • Leadership • Negotiation